

1 Ideal Over-the-Air Computation Baseline

Before introducing the practical impairments inherent to wireless channels, such as Inter-Symbol Interference (ISI), multipath fading, and time synchronization errors, it is essential to establish a baseline model of Over-the-Air (OTA) computation under ideal conditions. The foundational premise of OTA computing is its ability to reduce the communication latency of data aggregation from $\mathcal{O}(K)$ to $\mathcal{O}(1)$ by allowing K devices to transmit simultaneously [3].

In this idealized scenario, the wireless medium is assumed to be perfectly synchronized, lacking frequency-selective fading, and corrupted solely by Additive White Gaussian Noise (AWGN). This chapter aggregates the mathematical formulation, algorithmic MATLAB modeling, block-level Simulink architecture, and GNU Radio flowgraphs that demonstrate this ideal, unimpaired baseline. Establishing this benchmark is critical for quantifying the exact degradation caused by physical channel constraints in subsequent chapters.

1.a Mathematical Formulation of the Ideal Channel

Consider a distributed wireless sensor network consisting of K End Device (ED)s transmitting local analog values to a central Fusion Center (FC). Let $x_k(t)$ denote the continuous-time, real-valued signal transmitted by the k -th device. For analog OTA computation, uncoded Pulse Amplitude Modulation (PAM) is widely considered the optimal transmission scheme, as it linearly maps the sensor readings directly to the amplitude of the transmitted waveform [2], [4].

In an ideal Multiple Access Channel (MAC), we assume perfect Channel State Information (CSI) at the transmitters, which allows for flawless pre-equalization of phase shifts and path loss attenuation. Consequently, the electromagnetic waves superimpose perfectly at the receiving antenna of the FC. The received baseband signal $y(t)$ is mathematically expressed as the linear sum of the transmitted signals plus noise:

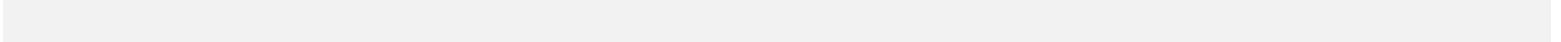
$$y(t) = \sum_{k=1}^K x_k(t) + n(t) \quad (1)$$

where $n(t) \sim \mathcal{N}(0, \sigma^2)$ represents the AWGN with zero mean and variance σ^2 . Because there is no multipath delay spread in this baseline model, the channel impulse response is effectively a Dirac delta function $h(t) = \delta(t)$. This implies that the current symbol is entirely unaffected by past or future symbols, resulting in zero ISI. To extract the arithmetic average of the K sensor readings, the FC simply applies a post-processing scaling factor of $1/K$ to the received aggregated signal $y(t)$.

1.b MATLAB Simulation and Code Analysis

To validate this superposition principle, the ideal environment can be modeled using MATLAB. The following script generates normally distributed simulated sensor readings, aligns them perfectly in time, and adds AWGN to emulate the ideal MAC.

Listing 1: MATLAB script for ideal OTA computation aggregation.

1 

```

9      % The Air acts as an ideal adder (MAC Superposition)
10     Ideal_Sum = sum(X, 1);
11
12     % Add AWGN
13     Received_Signal = awgn(Ideal_Sum, SNR_dB, 'measured');
14
15     % Calculate the Fusion Center Average
16     Estimated_Average = Received_Signal / K;
17
18     % Display Mean Squared Error (MSE)
19     MSE = immse(Estimated_Average, mean(X, 1)); fprintf('Ideal Baseline MSE
    at %d dB SNR: %f\n', SNR_dB, MSE);

```

While Listing 1 demonstrates the fundamental principle in its unadulterated form, practical wireless systems necessitate pulse shaping to satisfy spectral constraints and facilitate signal recovery. Consequently, a more physically realistic model incorporates matched Raised Cosine (RC) filtering at both the transmitter and receiver. This architectural refinement introduces additional degrees of freedom—namely, the Symbols per Sample (sps) parameter and the roll-off factor (α)—which materially influence system performance. The enhanced implementation is presented in Listing 2.

Listing 2: MATLAB script for OTA computation aggregation using the Root Raised Cosine (RRC) filter.

```

1      txUpSamped = upsample(X', sps_val); % phase may need to be set
2
3      % Raised Cosine Filter
4      rcFilter = rcosdesign(aleph, span, sps_val, "sqrt");
5
6      rx_agg_signal = zeros(N * sps_val + length(rcFilter) - 1, 1);
7
8      for k = 1:K
9          rx_agg_signal = rx_agg_signal + conv(txUpSamped(:,k), rcFilter(:));
10     end
11
12     Received_Signal = awgn(rx_agg_signal, current_snr, 'measured');
13
14     matchedSignal = conv(Received_Signal, flipud(rcFilter(:)), 'same');
15
16     rxMatched = downsample(matchedSignal, sps_val);
17
18     start_idx = floor(span/2); stop_idx = floor(span/2) + mod(span,2) + 1;
19
20     % Truncate edges to avoid filter transient errors
21     Estimated_Average = rxMatched(start_idx:end-stop_idx, :) / K;
22
23     % Calculate and store MSE in the matrix
24     mse_results(snr_idx, a_idx) = immse(Estimated_Average', mean(X, 1));

```

Computational execution of the aforementioned algorithms reveals a critical distinction between the idealized and practical

regimes. When temporal synchronization is mathematically perfect and no filtering is applied, the estimation accuracy is entirely governed by the noise floor; the Mean Squared Error (MSE) exhibits a direct proportionality to the AWGN variance and inverse quadratic scaling with the number of transmitting sensors. Mathematically, this relationship is expressed as $MSE \propto \sigma^2/K^2$, which demonstrates the fundamental advantage of cooperative multi-device aggregation [1], [3]. Conversely, when pulse shaping filters are incorporated into the transmission chain, the interplay between symbol timing parameters (sps) and spectral efficiency (Roll-off Factor (α)) introduces measurable deviation from the ideal theoretical bound.

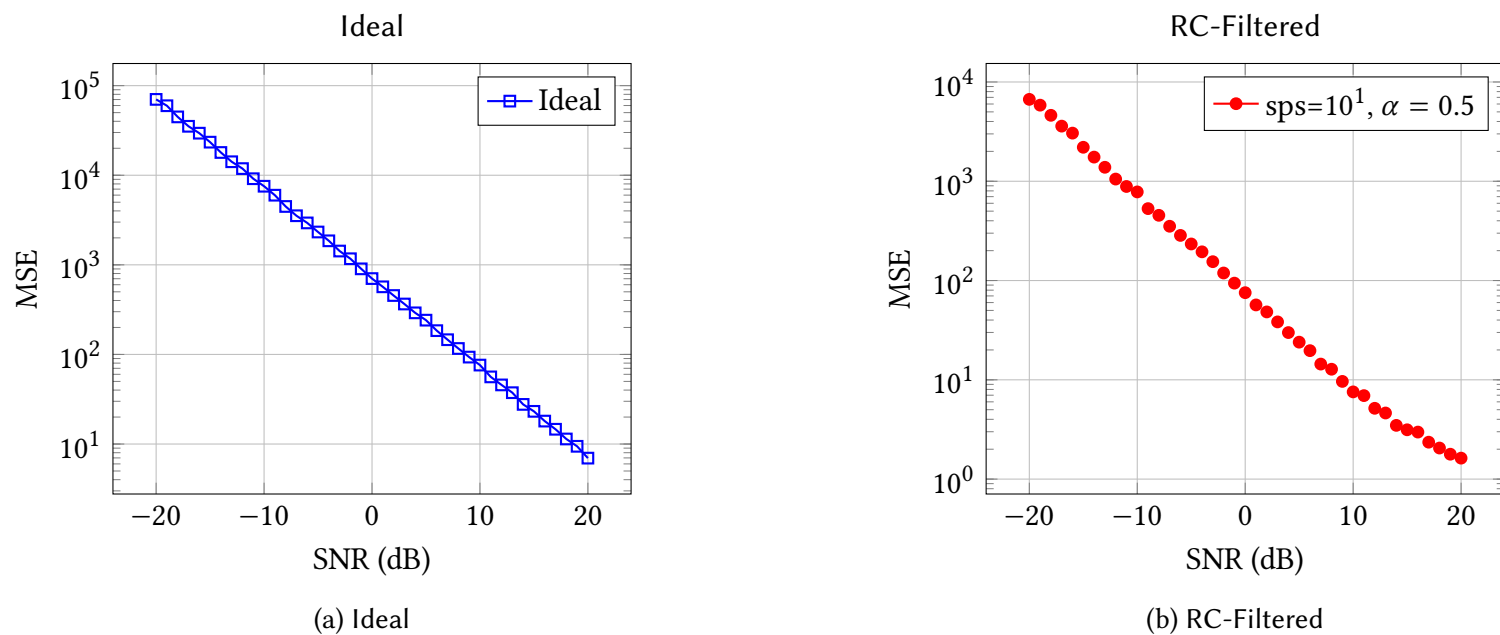


Figure 1: Comparison of MSE performance between the ideal baseline and the Raised Cosine filtered scenario.

Figure 1a presents the theoretical performance floor, wherein the MSE is a function solely of AWGN variance. In contrast, Figure 1b illustrates the performance envelope when matched Raised Cosine filters are deployed at both transmitter and receiver, with sps and α held constant. Despite the introduction of filtering, the channel remains free of multipath distortion and timing misalignment, ensuring that noise contributions remain the dominant performance-limiting factor.

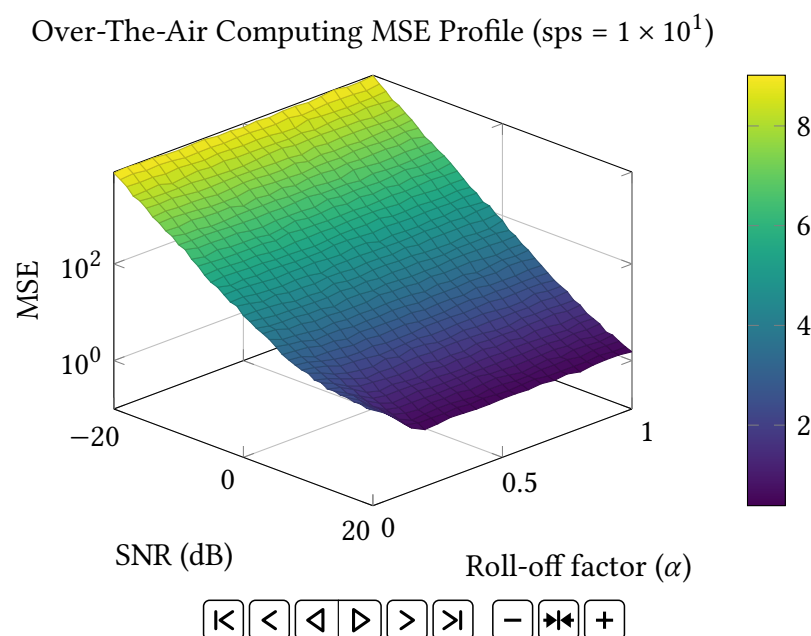


Figure 2: Interactive OTA computing animation demonstrating the architectural impact of Samples Per Symbol (sps) scaling.

A particularly noteworthy observation emerges from parametric analysis of the animation in Figure 2: while increasing sps generally yields marginal improvements in MSE performance across the SNR range examined, this enhancement asymptotically approaches a limit and does not constitute convergence toward the theoretical ideal baseline (absent pulse

shaping). This behavior underscores a fundamental trade-off inherent to practical pulse-shaping implementations: enhanced spectral efficiency through reduced sps incurs amplified inter-symbol interference, whereas elevated oversampling ratios improve timing robustness at the cost of increased computational complexity and memory consumption.

The eye and constellation diagrams provide valuable insights into the behavior of the ideal OTA system under different SPS configurations. As the number of samples per symbol increases, the eye opening widens, indicating reduced inter-symbol interference and improved signal integrity. Similarly, the constellation points become more distinct and well-separated, reflecting better performance in terms of error probability and overall system reliability.

The span parameter controls the width of the pulse shaping filter, and a larger span results in a more gradual transition band, leading to a flatter MSE curve across different roll-off factors.

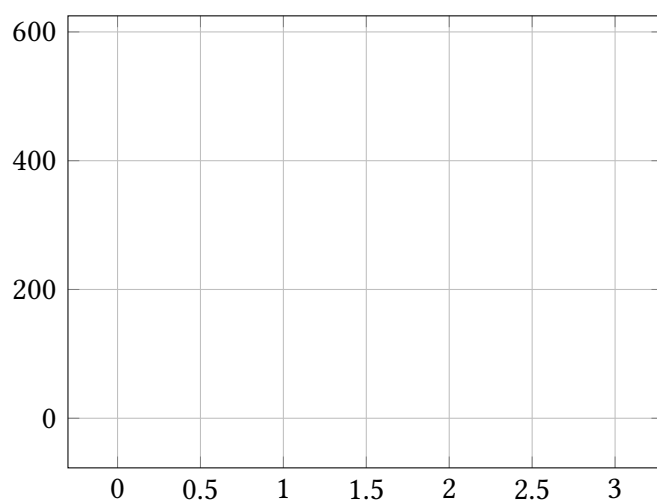
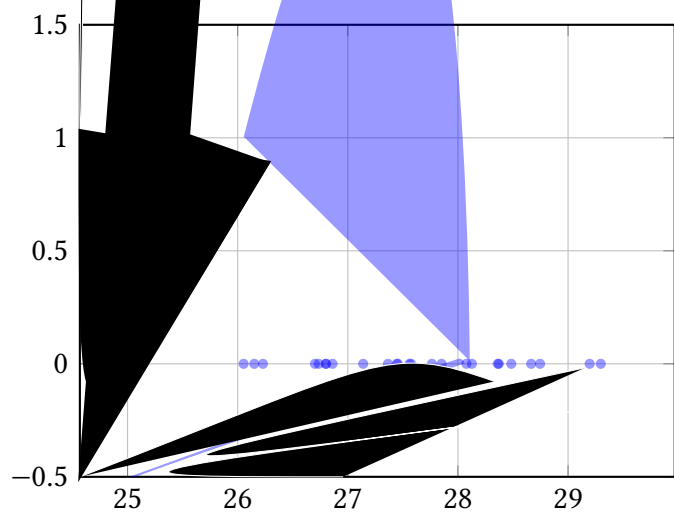
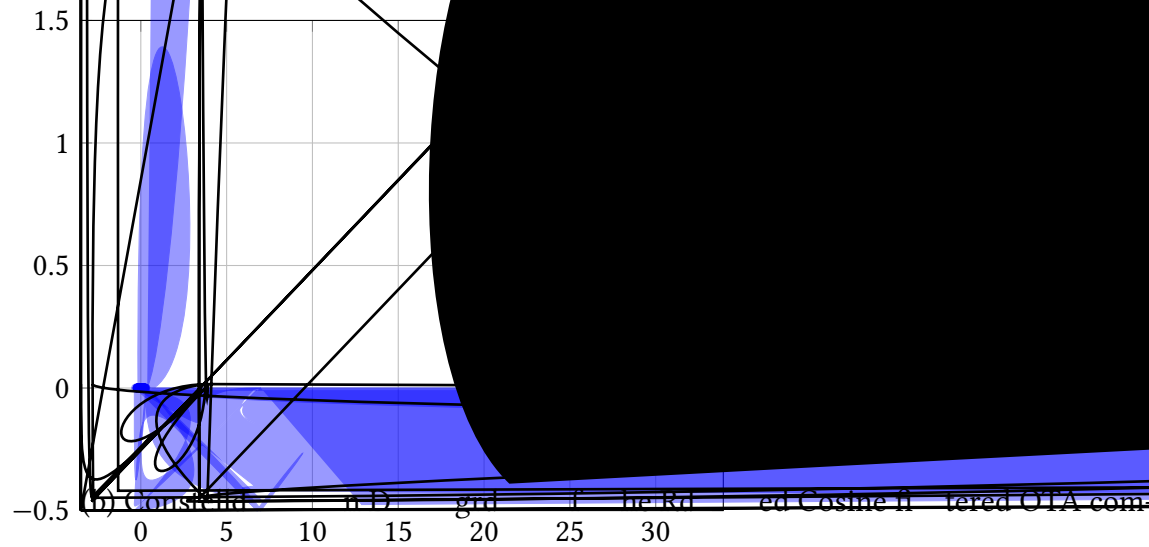




Figure 1: A block diagram representing the ideal Additive White Gaussian Noise (AWGN) channel.



The plot corresponding to the fusion center observation after the matched RRC filter shows a more compact and well-defined symbol distribution, indicating that the filtering stage has improved temporal alignment and reduced the impact of inter-symbol interference. In contrast, the constellation associated with the theoretical receiver model, where pulse shaping appears more dispersed and less structured, reflecting the absence of the temporal alignment provided by the matched filter. This comparison highlights the effectiveness of the matched filter in preserving symbol integrity while mitigating ISI.



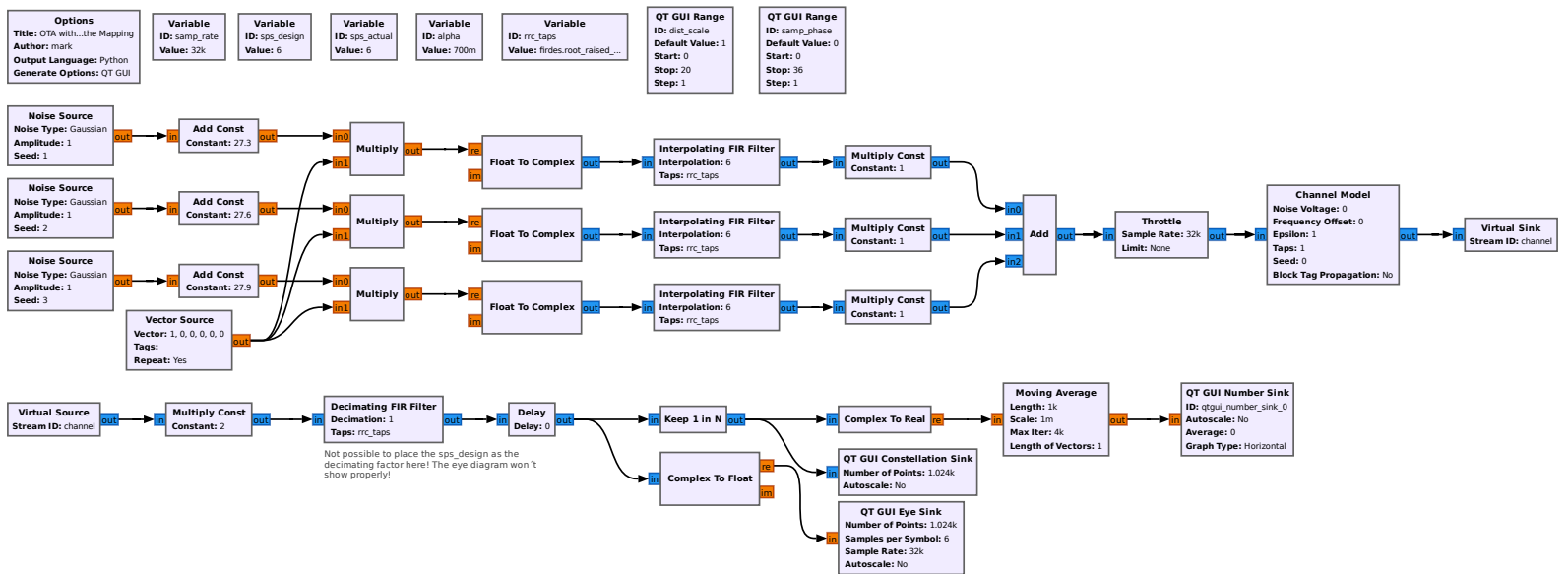


Figure 10: GNU Radio flowgraph for the ideal, Nyquist-compliant scenario.

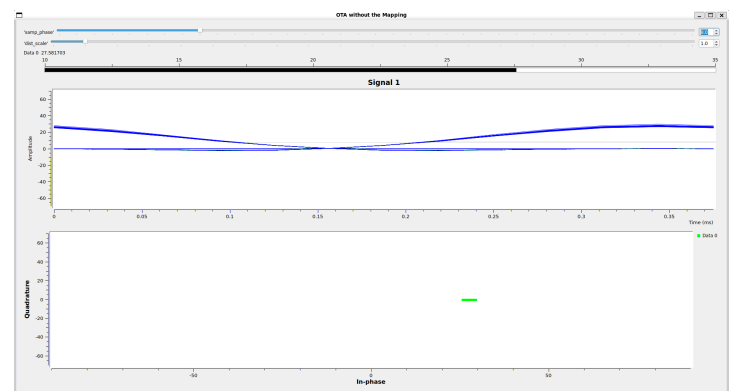
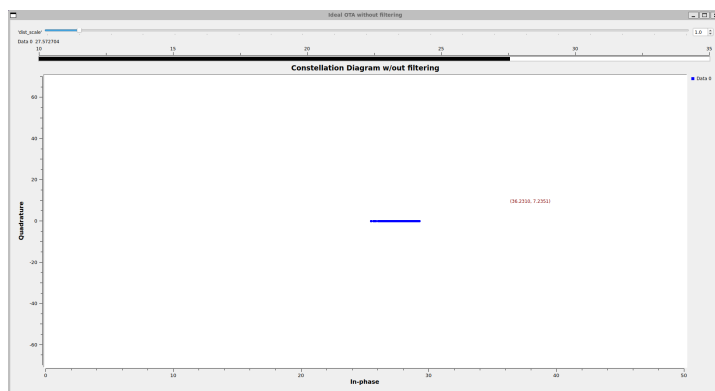


Figure 11: GNU Radio flowgraph for the ideal, Nyquist-compliant scenario.

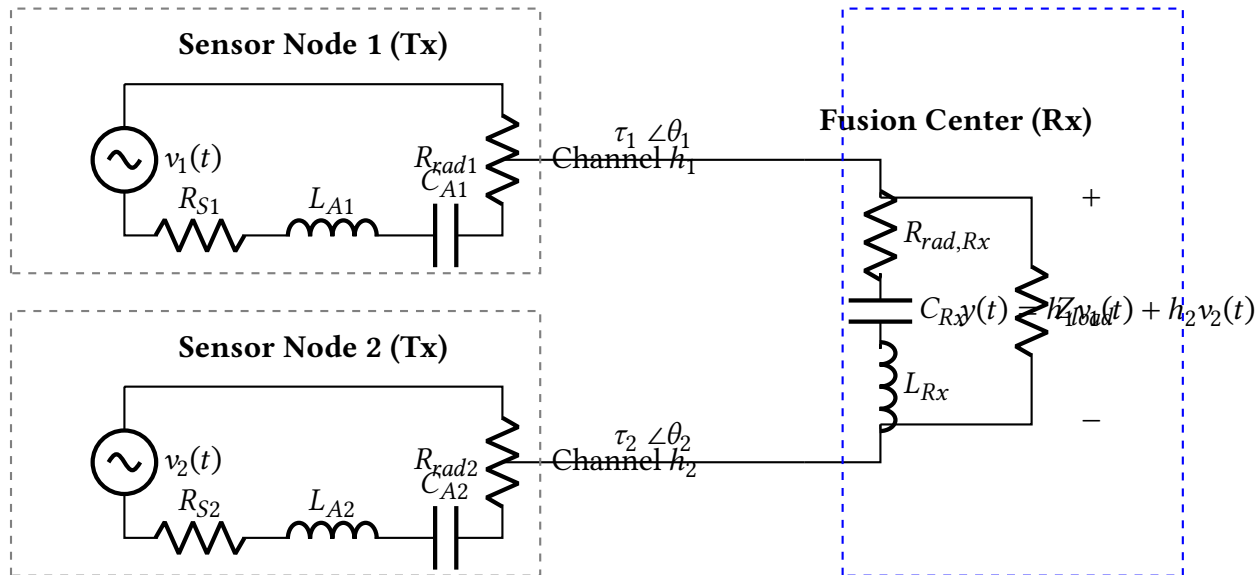


Figure 12: Circuit equivalent of the ideal OTA computation network. Each sensor node is modeled as a voltage source with internal impedance, and the fusion center aggregates the received signals through a common load. The wireless channels are represented as transmission lines with associated delays and phase shifts, capturing the effects of propagation in a simplified manner.

introduces additional practical considerations, such as real-time processing constraints and hardware-induced latency. These factors may influence the observed performance metrics, particularly in scenarios where the system is subjected to dynamic channel conditions or when operating near the limits of the hardware's processing capabilities. Therefore, while the idealized models provide a valuable theoretical benchmark, real-world deployments must account for these additional complexities to ensure robust and reliable operation.

The circuit equivalent of the ideal OTA computation network is depicted in Figure 12. This representation abstracts the wireless channel into a simplified electrical model, where each sensor's transmission is treated as a voltage source, and the fusion center aggregates these signals through a common load. The circuit

References

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